

Next, I discussed how class variables are different from instance variables. Every object instantiated from a class gets its own set of all the instance variables in the class. However, there is only one of each class variable, and that is accessible by all objects created from the class. Class variables are often used as constants or counters, or for loading something large and making it available to all objects instantiated from the class.

To put a number of techniques and concepts together, we built a balloon-popping game and organized it very efficiently. We had one file that contained only constants used by other files. The main code consisted of the main loop and a status display, and the balloon manager did the work of managing the objects. Such a division of labor allows for splitting up the game into smaller, logical pieces. The role of each part is well defined, making the overall program more manageable.

Finally, I explained how a technique called slots allows for a memory-efficient representation of instance variables.